SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Database Management Systems
COURSE CODE: CSA0509

COURSE OUTCOME COVERED

CO3: *Analyze relational databases using functional dependencies, normalization (1NF to 5NF), and key constraints to identify redundancy and ensure data integrity.* (BL2, BL3)

Activity Tool : MCQ Test (Low)
Assessment Tool Weightage: 30%

QUESTIONS Total Marks: 50

Q.No	Question	Bloom's Level	Marks
1	Which of the following is true about 1NF? (a) Allows multi-valued attributes (b) Eliminates partial dependencies (c) All attributes must be atomic (d) Eliminates transitive dependencies	BL2 – Understand	5
2	A relation is in 2NF if it is in 1NF and: (a) Has no transitive dependencies (b) Has no partial dependencies on primary key (c) Every attribute is a prime attribute (d) Has no multi-valued dependencies	BL2 – Understand	5
3	Which normal form deals with multi-valued dependencies? (a) 2NF (b) 3NF (c) BCNF (d) 4NF	BL2 – Understand	5
4	BCNF is a stricter version of: (a) 1NF (b) 2NF (c) 3NF (d) 4NF	BL2 – Understand	5
5	Functional dependency X → Y means: (a) Y determines X (b) X determines Y (c) X and Y are unrelated (d) Both determine each other	BL2 – Understand	5
6	Which of the following anomaly is NOT associated with unnormalized relations? (a) Insertion anomaly (b) Deletion anomaly (c) Update anomaly (d) Retrieval anomaly	BL2 – Understand	5
7	A superkey is: (a) A minimal set of attributes that uniquely identifies tuples (b) Any set of attributes that uniquely identifies tuples (c) An attribute that depends on a non-prime attribute (d) A foreign key reference	BL2 – Understand	5

8	If R(A,B,C) has FD: A→B and B→C, then C is: (a) Partially dependent on A (b) Fully dependent on A (c) Transitively dependent on A (d) Independently determined	BL2 – Understand	**5**
9	5NF is also known as: (a) Domain-Key Normal Form (b) Boyce-Codd Normal Form (c) Project-Join Normal Form (d) Element Normal Form	BL2 – Understand	**5**
10	Lossless-join decomposition ensures: (a) No data is lost during decomposition (b) All FDs are preserved (c) The relation is in BCNF (d) No null values exist	BL2 – Understand	**5**

Code of Conduct:

*I **_Monika.R(192324281)**__(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Monika.R

RUBRICS FOR CALCULATION:

MCQ					
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions

CSA0509 – DATABASE MANAGEMENT SYSTEM

CO3-AT3

MULTIPLE CHOICE QUESTIONS

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Submitted To:	Submitted on:
Dr. R Kesavan	12-08-2026

1. Which of the following is true about 1NF?

A) Allows multi-valued attributes

B) Eliminates transitive dependencies

C) All attributes must be atomic

D) Eliminates partial dependencies

Answer: C) All attributes must be atomic

Explanation: In 1NF, each cell must contain a **single atomic value**.

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2. A relation is in 2NF if it is in 1NF and:

A) Has no transitive dependencies

B) Has no partial dependencies on primary key

C) Every attribute is a prime attribute

D) Has no multi-valued dependencies

Answer: B) Has no partial dependencies on primary key

Explanation: 2NF removes **partial dependency**, where a non-key attribute depends on only part of a composite key.

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3. Which normal form deals with multi-valued dependencies?

A) 2NF

B) 3NF

C) BCNF

D) 4NF

Answer: D) 4NF

Explanation: Fourth Normal Form (4NF) handles multivalued dependencies.

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4. BCNF is a stricter version of:

A) 1NF

B) 2NF

C) 3NF

D) 4NF

Answer: C) 3NF

Explanation: BCNF is a stronger form of **Third Normal Form (3NF)**.

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5. Functional dependency X → Y means:

A) Y determines X

B) X determines Y

C) X and Y are unrelated

D) Both determine each other

Answer: B) X determines Y

Explanation: In **X → Y**, knowing X allows us to determine Y.

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6. Which of the following anomaly is NOT associated with unnormalized relations?

A) Insertion anomaly

B) Update anomaly

C) Deletion anomaly

D) Retrieval anomaly

Answer: D) Retrieval anomaly

Explanation: The three standard anomalies are **insertion, update, and deletion**.

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7. A superkey is:

A) A minimal set of attributes that uniquely identifies tuples

B) Any superset of an attribute that uniquely identifies tuples

C) An attribute that depends on a non-prime attribute

D) A foreign key reference

Answer: B) Any superset of an attribute that uniquely identifies tuples

Explanation: A **superkey** uniquely identifies a row. A candidate key is a **minimal superkey**.

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8. If R(A,B,C) has FD: A → B and B → C, then C is:

A) Partially dependent on A

B) Fully dependent on A

C) Transitively dependent on A

D) Independent of A

Answer: C) Transitively dependent on A

Explanation: A → B and **B → C**, so **A → C indirectly**. This is transitive dependency.

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9. 5NF is also known as:

A) Domain-Key Normal Form

B) Boyce-Codd Normal Form

C) Project-Join Normal Form

D) Element Normal Form

Answer: C) Project-Join Normal Form

Explanation: 5NF = Project-Join Normal Form (PJ/NF).

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10. Lossless-join decomposition ensures:

A) No data is lost during decomposition

B) All FDs are preserved

C) The relation is in BCNF

D) No null values exist

Answer: A) No data is lost during decomposition

Explanation: In a **lossless decomposition**, the original relation can be reconstructed without losing information.